TECHNICAL SYSTEM ARCHITECTURE & SPECIFICATION SHEET

Fast X Logistics Engine Specification

An industrial-grade transaction, telemetry, and operational engine engineered for secure dispatch coordination, cash protection, and high-frequency parcel routing.

AUTHOR / ARCHITECT

Dunsimi

VERSION

v1.2 (Enterprise-Ready)

SECURITY LEVEL

Strictly Confidential

DEPLOYMENT ROADMAP

Web PWA to Native Mobile

1. Strategic Deployment & Scaling Roadmap

To maximize early operational efficiency, mitigate infrastructure complexity, and align capital deployment with transaction velocity, Fast X will be launched in a Multi-Phase Deployment Roadmap. This allows the business to scale seamlessly from day-one without high initial operational overheads.

Phase 1: Progressive Responsive Web Application (Launch State)

- Architecture: Monolithic or decoupled responsive web application (HTML5, Vanilla CSS, JS/TypeScript) optimized for mobile browser runtimes (Chrome Mobile, Safari iOS).
- Justification: Zero friction onboarding for casual customers and B2B vendors who can access the system instantly without downloading packages. Highly responsive layout architectures mimic native applications while keeping the codebase unified under a single development team.
- Rider Constraints: Addresses mobile browser limitations (aggressive JS sleeps, GPS throttling) through robust front-end telemetry caching.

Phase 2: Hybrid Native Wrapper Framework (Transition State)

- Architecture: Packaging the existing web codebase into hybrid frameworks (e.g., Capacitor, Cordova, or Flutter Webview wrappers).
- Objective: Establish presence in the Google Play Store and Apple App Store while maintaining a single codebase. Provides access to native device features (persistent background location services, push notification channels, and direct hardware API integration).

Phase 3: Fully Native / Cross-Platform Mobile Applications (Scale State)

- Architecture: Rewriting the client interfaces into highly performant native frameworks (React Native, Flutter, Swift, and Kotlin).
- Objective: Deploy dedicated high-frequency applications optimized for riders (with hardware-level GPS tracking and background geolocation daemons) and high-frequency business vendors.

2. Comprehensive Page Directory

The Fast X ecosystem consists of four distinct portals tailored to specific user archetypes. The web application will launch with the following structural layout:

2.1 Public Pages (No Authentication Required)

- Corporate Landing Page (/): High-premium dark-mode UI detailing primary CTAs for bookings, vendors, and couriers.
- Dynamic Logistics Quote Estimator (/quote): Interactive pricing wizard mapping pickup/dropoff States and LGAs natively with weight class selection.
- Universal Tracking Portal (/track): Search console utilizing client-side linear interpolation to slide rider position markers smoothly along vectors.
- Rider Onboarding Hub (/join-courier): Detailed portal containing earning structures, commission splits, and registration guidelines.
- B2B Vendor Marketing Hub (/vendors): Core product highlights for high-frequency e-commerce stores (batch spreadsheets uploads, COD Escrows).

2.2 Customer & B2B Vendor Portal

- Casual Customer Dashboard (/dashboard/customer/casual): Passwordless OTP magic links via WhatsApp/SMS detailing booking histories.
- B2B Vendor Workstation (/dashboard/customer/vendor): Bulk Excel/CSV parser engine, keyboard-navigable rapid tabular entry screen, Kanban-style Active Order Pipeline tracker, and a Dual-Balance COD Escrow Vault.

2.3 Rider / Courier & Admin Portals

- Rider Onboarding & Compliance Funnel (/dashboard/rider/onboarding): Stateful 3-step registration (WhatsApp verification, Paystack name validation, NIN/License upload).
- Active Courier Workspace (/dashboard/rider/workspace): Atomic Job Marketplace, Wallet Ledgers, and stateful PIN validation execution grids.
- Admin Dispatch Board (/dashboard/admin/dispatch): Ops dispatch control, chat systems, manual payment confirmation panel.
- Super Admin Governance Panel (/dashboard/admin/governance): RBAC-enforced matrix price editing, staff creation, payout controls, and immutable write-once System Activity Logs.

3. Core Telemetry Loop & Security Matrices

3.1 Hybrid Telemetry Engine Specifications

1. Position Throttling: Coordinates are only sent to the server if the rider crosses a 50-meter vector threshold OR 30 seconds have elapsed.
2. Offline Resiliency: Captures location breadcrumbs locally inside the browser's IndexedDB / localStorage during network drops, running batch writes to the server upon reconnection.
3. High-Throughput Memory Cache: Geolocation coordinates bypass SQL databases and are written directly to high-speed in-memory layers (Redis/Memcached). Transient coordinates are completely deleted upon delivery completion.

3.2 Transaction Checkout & Resiliency Matrices

Subsystem Component	Technical Safeguard Enforced	Failure State Resiliency Pattern
Paystack Webhooks	Payload validation & Idempotency Locks.	Ignore retry webhooks, prevent duplicate dispatches.
WhatsApp Checkout	Order ID tracking with deep-linked parameters.	Stores booking as pending bank transfer check.
Cash on Delivery	Collection_Type: CASH with rider wallet debt.	Lock rider requests until balance matched.
Telemetry Failure	Automated Heartbeat Monitor cron-daemon.	In-transit drops flag high-priority dispatcher alert.
Delivery Handshake	Two-Way Verification (Pickup & Delivery PINs).	Escrow locked until recipient PIN validated.